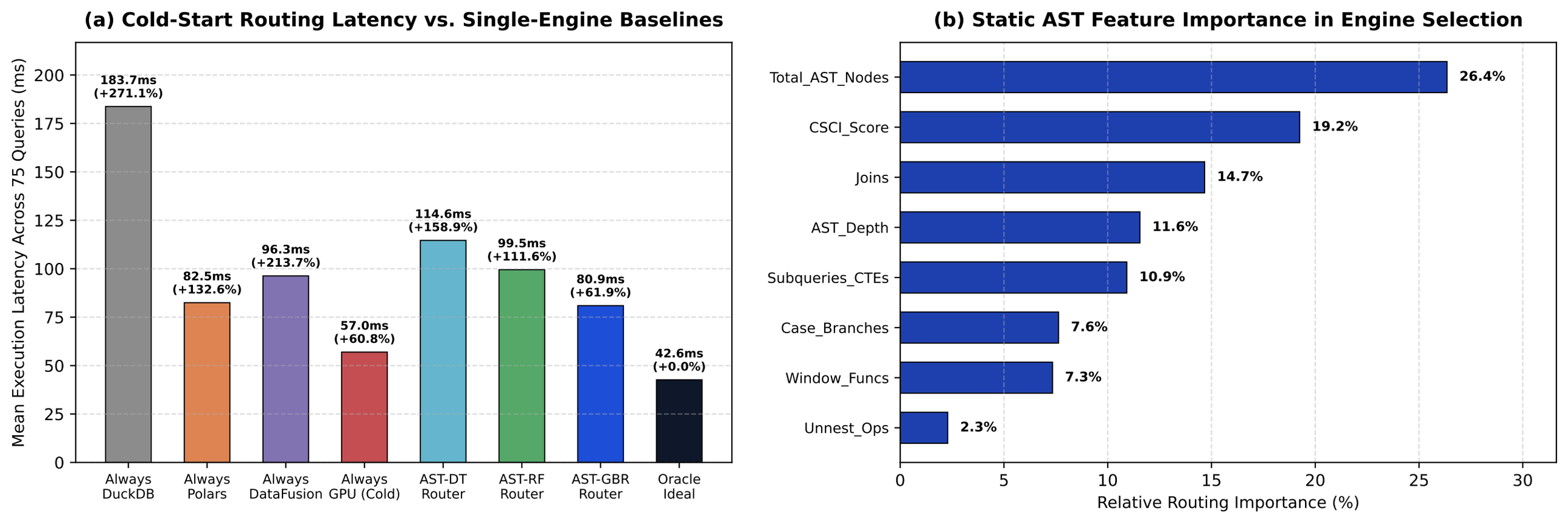
(a) Cold-Start Routing Latency vs. Single-Engine Baselines
Mean Execution Latency Across 75 Queries (ms)
0
25
50
75
100
125
150
175
200
183.7ms
(+271.1%)
82.5ms
(+132.6%)
96.3ms
(+213.7%)
57.0ms
(+60.8%)
114.6ms
(+158.9%)
99.5ms
(+111.6%)
80.9ms
(+61.9%)
42.6ms
(+0.0%)
Always DuckDB
Always Polars
Always DataFusion
Always GPU (Cold)
AST-DT Router
AST-RF Router
AST-GBR Router
Oracle Ideal
(b) Static AST Feature Importance in Engine Selection
Total_AST_Nodes
CSCI_Score
Joins
AST_Depth
Subqueries_CTEs
Case_Branches
Window_Funcs
Unnest_Ops
26.4%
19.2%
14.7%
11.6%
10.9%
7.6%
7.3%
2.3%
0
5
10
15
20
25
30
Relative Routing Importance (%)